

A Multi-Objective Optimization Scheme for Job Scheduling in Sustainable Cloud Data Centers

Kuljeet Kaur^{ID}, *Member, IEEE*, Sahil Garg^{ID}, *Member, IEEE*,
Gagangeet Singh Aujla^{ID}, *Senior Member, IEEE*,
Neeraj Kumar^{ID}, *Senior Member, IEEE*, and Albert Y. Zomaya^{ID}, *Fellow, IEEE*

Abstract—For a number of years, due to an exponential increase in the demand for an eco-friendly environment, there has been a rapid increase in the green city revolution across the globe. Subsequently, load shifting of major energy consumers from conventional power grids to renewable energy sources (RES) has become inevitable. Towards this end, cloud data centers (DCs) have emerged as significant consumers of energy that solely rely on power grids to fuel their day-to-day operations. Nevertheless, their energy consumption has increased significantly which in turn has substantially raised the global carbon footprint rate. These challenges can be best addressed by the judicious utilization of RES which have well established advantages like reduced operational costs and carbon emissions. Keeping in view of the above facts, the ultimate goal of the proposed work is to design a comprehensive workload classification; and job scheduling and Virtual machine placement architecture for cloud DCs powered by RES and power grids. For this, a multi-objective optimization scheme is proposed which operates in two phases. In phase I, a *random forest-based wrapper scheme* known as Boruta, is used for relevant feature set selection for the incoming workload. This is followed by classification of the workload using a locality sensitive hashing-based support vector machines approach. In phase II, a multi-objective optimization problem for job scheduling and VM placement is formulated with respect to parameters such as service level agreement (SLA), energy cost, carbon footprint rate (CFR), and availability of RES. It is further solved using an enhanced heuristic approach based on a greedy strategy. Our experimental evaluations show an average improvement of approximately 31 percent in energy utilization, 28 percent in energy cost, and 36 percent in CFR, with a slight degradation in SLA assurance (about 2 percent) compared with the existing schemes.

Index Terms—Classification, cloud data centers, multi-objective optimization, sustainability, VM placement

1 INTRODUCTION

CLOUD computing (CC) is one of the most powerful technologies of the modern era for providing on-demand services to end users as per their demands. In order to cater to the service requirements of the intelligent end user domain, CC has evolved as a ubiquitous platform by virtue of virtualization. But, adherence to the service-level agreement (SLA) is a challenging task while providing parallel and distributed resources to end users. In CC, virtual resources are sliced over physical servers to handle incoming workloads [1]. For this purpose, massive data centers (DCs) are deployed by different service providers across the globe. With a rapid growth in the number of smart users, the need for different types of cloud

resources has increased exponentially [2]. In order to handle millions of job requests, DCs deployed across the globe have witnessed an exponential growth in terms of number, size, and functionalities. For example, to provide cloud resources for efficient provision of on-line activities such as-email, social media, and e-commerce across the US, nearly three million DCs are deployed [3]. However, with the advent of the latest technologies such as industrial internet of things, smart healthcare, and intelligent transportation; the existing computing capabilities of DCs would soon be insufficient. According to [4] (2014), only 8.5 percent of global DCs had enough resources to cater for the increasing end user demands beyond 2015 and the years to come. It has been projected that 75 percent of DCs deployed by various cloud service providers (CSPs) are expected to expand both vertically and horizontally. By 2020, such an expansion is expected to double the size of DCs relative to 2010.

For routine functioning (such as resource scheduling and management, computations etc.) of DCs, a huge amount of operational (OPEX) cost is incurred [5], [6]. Energy consumption is one of the major components of DC's overall OPEX cost. According to Mostelic and Brandic [4], energy consumption of DCs is expected to increase from 70 billion kWh (2000) to 1000 billion kWh (2020) [4]. Subsequently, DC's OPEX cost would inevitably increase manifold. According to environment protection agency [7], \$4.5 billion is spent annually on the energy consumption of DCs located in the US alone. Moreover, a report in 2012 [3] stated that DCs located in the US consume energy equivalent to the

- K. Kaur and S. Garg are with Department of Electrical Engineering, École de Technologie Supérieure, Université du Québec, Montréal, QC H3C 1K3, Canada. E-mail: {kuljeet.kaur, sahil.garg}@ieee.org.
- G.S. Aujla is with the Newcastle University, Newcastle upon Tyne NE1 7RU, United Kingdom and also with the Chandigarh University, Mohali, (Punjab), India. E-mail: gagi_aujla82@yahoo.com.
- N. Kumar is with the Department of Computer Science and Information Engineering, Asia University, Taiwan, and with the King Abdulaziz University, Jeddah, Saudi Arabia, and also with the Department of Computer Science & Engineering, Thapar Institute of Engineering and Technology (Deemed to be University), Patiala, Punjab 147001, India. E-mail: neeraj.kumar@thapar.edu.
- A.Y. Zomaya is with the School of Computer Science, University of Sydney, NSW 2006, Australia. E-mail: albert.zomaya@sydney.edu.au.

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(Corresponding author: Neeraj Kumar.)

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generation capacity of 34 coal-powered 500 MW power plants. However, these are expected to double in order to meet the growing energy demand of DCs deployed in the US alone. So, it is evident from these reports that in order to cope with the growing energy demand of DCs handling millions of job request, the power grid to which they are connected has to bear additional load. In such a case, additional coal-powered power plants are required which may escalate global carbon emissions.

According to a NRDC report in 2013 [3], in the US alone, DCs contributed 97 MMT of carbon which is expected to reach 147 MMT by 2020. So, the amount of carbon and operational cost related to the energy consumption of DCs represent a serious threat to the vision of future green cities. Hence, to deal with such a situation, CSPs are using energy-efficient techniques and technologies such as virtualization, consolidation, container migration and software defined networking in order to handle millions of job requests [8], [9], [10], [11], [12], [13], [14]. However, provision of reliable cloud services with desirable quality of service (QoS) and SLA adherence is one of the major challenges for CSPs [15]. For this reason, the focus of CSPs has shifted from the usage of energy-efficient techniques towards sustainable computing. Major CSPs such as Apple, Amazon, and Google are already on the path of designing sustainable DCs [16]. So, moving in this direction, deploying renewable energy sources (RES) as an alternate source of energy to power DCs is the need of hour. Hence, using RES along with energy-efficient techniques can provide many benefits such as reduction in energy consumption, energy cost, carbon footprints and additional load on grid [17].

Belaglazov et al. [11] highlighted that green cloud computing is one of the best solutions to tackle the issue of increasing energy consumption of DCs. However, to achieve this goal, coordination among energy-efficient techniques (resource management, job scheduling, etc) and energy management approaches is essential. For this purpose, a few existing proposals presented some solutions to design sustainable DCs using RES. For example, Yu et al. [9] proposed an optimization technique for energy management of DCs to handle power outages using RES. Polverini et al. [18] presented a thermal-aware job scheduling scheme to minimize energy consumption of geo-distributed DCs. In other work, Khosrovi et al. [13], proposed a VM placement scheme with respect to carbon footprint rates and power usage effectiveness. Guo et al. [19] presented a thermal storage-based energy and network aware workload management scheme for sustainable DCs. Chen et al. [20] proposed a workload and energy management scheme for geo-distributed DCs to achieve sustainability using distributed energy storage and RES. Similarly, Auja et al. [3], [21] designed an energy management scheme for sustainability of DCs using electric vehicles. In another work, authors [14] formulated a Stackelberg game for sustainable resource allocation in geo-distributed DCs using RES.

On the similar lines, authors in [22] proposed a time-aware task scheduling scheme for Green DCs using hybrid meta-heuristic approaches. The designed scheme was named TATS and was designed to maximize the profit of Green DCs while keeping the delay bound of submitted jobs into account. On the contrary, the significant concentration of the work

proposed in [23] has been on revenue maximization of cloud DCs. To accomplish the same, the authors suggested an admission control method followed by the workload scheduling problem. The cost parameter considered in this work included the expenditure incurred by the ISP bandwidth and energy consumption of cloud DCs. Likewise, the work in [24] aims at maximizing the overall profit of geographically distributed DCs. The work also takes in accounts the following aspects while scheduling the incoming workload to different DCs, i.e., self-sustainability & green energy utilization, electricity price variability and the SLA of incoming jobs. It was achieved by the authors by formulating an optimization-based workload distribution scheme that relies on the accuracy of G/D/1 queue. Polverini et al. [25] explored the job scheduling problem across geographically distributed data centers. The authors proposed an Online Algorithm referred as GreFar for the same which took in consideration the following key objectives: energy cost, fairness and queuing delay along with the server inlet temperature constraints.

However, in the quest of sustainable environment and computing, it is essential for the cloud computing fraternity to give utmost importance to the use of RES and reduce the emission carbon footprint rate (CFR). This demands designing of intelligent energy-aware scheduling strategies using the dominant paradigm of virtualization to achieve the above-mentioned objectives. But to the above best of the author's knowledge, the existing research proposals have not explored these objectives in conjunction to pave the way to sustainable computing. For instance, the majority of the current state-of-the-art has not considered the CFR challenges [22], [23], [24], [25], [26], [27]. While the works in [22], [23], [24] did not use the virtualization scheme to achieve additional benefits of tasks scheduling and consolidation, further, the use of RES is not explored in various articles [23], [25] up to its full potential. The illustrative comparison of the proposed scheme with some of the existing state-of-the-art has been summarized in Table 1.

1.1 Research Contributions of This Work

The research contributions of the work are manifold and can be segregated into two phases. In the first phase, the incoming workload is classified by relevant feature set obtained using Boruta. Once the workload is classified, it is then scheduled on the available array of servers based on the proposed Multi-Objective Optimization Problem (MOOP); this constitutes the next phase. The coupling of the two phases in the context of green-energy supported DCs is a novel research contribution that has not yet been explored up to its full potential. Inspired by the above-discussed facts, the following contributions are presented in this paper.

- An *all-relevant random forest-based wrapper scheme* namely *Boruta* is used to select feature sets for classification of incoming workloads at geo-distributed DCs.
- Using the selected feature set, a workload classification scheme using support vector machines (SVM) is designed. To optimize this scheme, a scalable locality sensitivity hashing (LSH) approach is used.
- A multi-objective optimization problem is formulated according to different parameters such as SLA adherence, energy cost, carbon footprint rate, and RES sustainability.

TABLE 1
Relative Comparison of the Proposed Scheme With the Current State-of-the-Art

Scheme	Feature Selection	Classification	MOOOP	SLA+QoS	Energy Cost	CFR	RES	VMs	Remarks
Yuan <i>et al.</i> [22]	×	×	✓	✓	✓	×	✓	×	Proposed a time-aware task scheduling strategy for Green DCs using hybrid meta-heuristic approach
Yuan <i>et al.</i> [23]	×	✓	✓	✓	✓	×	×	×	The major concentration has been on revenue maximization of cloud DCs.
Yuan <i>et al.</i> [26]	×	×	✓	✓	✓	×	×	×	A temporal task scheduling algorithm based on MILP was proposed to minimize the cost of private cloud DCs.
Zhang & Zhou [27]	×	✓	×	✓	×	×	×	✓	Bayes classifier is used in first stage and single objective optimization approach is employed in the second stage for task scheduling
Polverini <i>et al.</i> [25]	×	×	✓	✓	✓	×	×	✓	Proposed an Online Algorithm for job scheduling while reducing the impact of maximum server inlet temperature
Kiani & Ansari [24]	×	✓	✓	✓	✓	×	×	×	G/D/1 queuing is used for workload distribution and job scheduling is done across datacenters.
Ours	✓	✓	✓	✓	✓	✓	✓	✓	Employees Boruta for feature selection, SVM+LSH for job classification and Greedy Strategy for job scheduling

- Finally, an enhanced heuristic approach for job scheduling and VM placement is presented.

1.2 Organization

The rest of this paper is organized as follows. Section 2 describes the System model and problem formulation. Section 3 presents the Boruta algorithm for feature selection and a LSH-based SVM approach for workload classification. In Section 4, a MOOP for sustainability of DCs is proposed followed by an enhanced heuristic approach for job scheduling and VM placement algorithm. In Section 5 results are discussed. Finally, Section 6 concludes the article.

2 SYSTEM MODEL

In this section, the system architecture of the proposed scheme is discussed along with the working of the underlying phases for job classification and scheduling.

The system under consideration is a typical IaaS cloud provider system with the provision of scheduling the

incoming jobs on the available array of servers using VMs. Hence, the overall agenda of the work is to design an effective job scheduling and VM placement approach while attaining a desirable trade-off between the competing objectives, namely service level agreement, energy cost, CFR, and availability of RES.

The overall working architecture of the proposed scheme is depicted in Fig. 1. In the proposed system architecture, n geo-distributed DCs powered by RES and grid (through a battery energy storage system) have been considered. The provisioning of the incoming workload to these servers is achieved using virtualization across two phases. In the first phase, the incoming workload (W) is classified into different queues ($Q_1, Q_2, \dots, Q_n$) using the proposed LSH-based SVM classifier. The classification is initially made by the selection of the relevant feature set using Boruta. In the next phase, the classified workload is scheduled among DCs by a cloud broker which comprises of job scheduler, VM placement manager, etc. The job scheduling and VM placement approach adopted in this phase maintains the trade-off with the competing functions using MOOP. The formulated

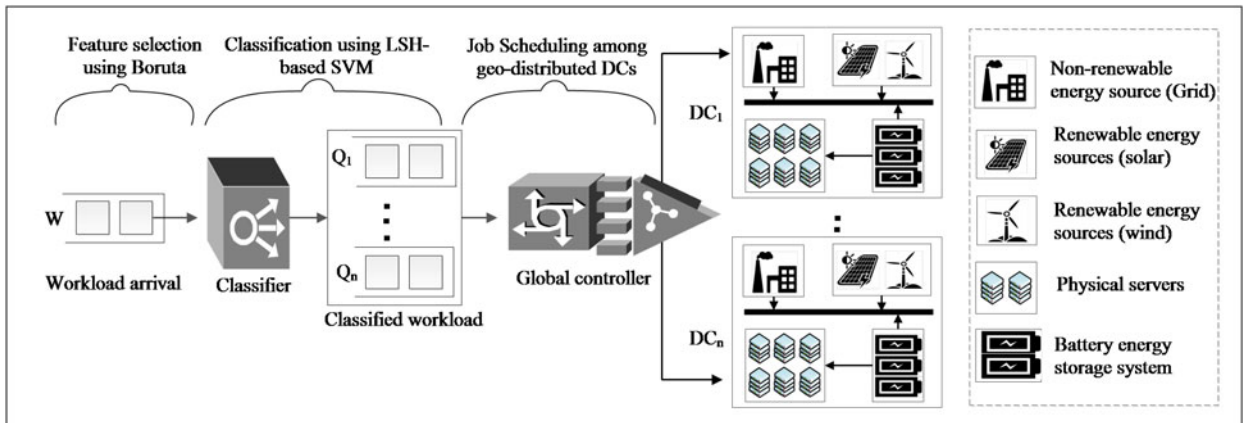


Fig. 1. System architecture of the proposed scheme.

MOOP is solved using a heuristic approach based on a greedy strategy keeping in mind the complexity of the underlying system. The working of the above mentioned phases is elaborated in the upcoming sections.

3 PHASE I: FEATURE SELECTION AND CLASSIFICATION

In this phase, the incoming workload is initially subjected to a Boruta-based feature selection strategy for selecting the relevant features. Using the selected features, the cloud broker then performs LSH-based SVM classification. The detailed information is provided below.

3.1 Boruta-Based Feature Selection

Feature selection is a crucial step in predictive modelling. It helps to build predictive models free from correlated variables, biases and unwanted noise [28]. In such an environment, Random Forest classifiers are predominately used due to their advantages such as relatively quick convergence, ability to operate without tuning of parameters, numerical estimate of the feature importance, and its applicability for both linear and non-linear relationships among different data items. But, Random Forest suffers from misleading impact of fluctuations and correlations. So, in this paper, the Boruta algorithm [29] is used to provide unbiased and stable selection of important and non-important attributes from an information system. Due to the iterative construction, Boruta can deal with both the fluctuating nature of a random forest importance measure and the interactions between attributes. It is defined as below:

Definition 1 (Boruta-based Feature Selection). *Boruta is an all-relevant feature selection algorithm which is based on the foundations of the random forest classifier, i.e., a predictor variable f_i is said to be relevant to f_j , if f_i and f_j are not conditionally independent.*

The algorithm for feature selection using Boruta is described in Algorithm 1. This algorithm operates as follows: In an initial stage, an extended system is built by creating shadow attributes (var_f^{shad}) for all features $f_t \in F_S$ where these are shuffled in order to remove the correlations and biases among them. A random forest classifier (RF_Classifier) is then trained on the extended version of information system to evaluate the importance of each feature (var_{imp}). Here, classification is performed on the basis of decision trees which are independently developed on different bagging samples of the training set. This provides an intrinsic measure of the importance of each randomized attribute known as the z-score. This score is not a statistically significant solution for the significance of the feature, thus the procedure is repeated several times. During each iteration, maximal importance to the random set of attributes (max_{z_score}) is computed by comparing it with a set of real attributes. A statistical test is then performed for all the attributes to get stability in the test where irrelevant features are progressively eliminated. Hence, only those features whose importance is higher than that of the max_{z_score} are deemed "Important", and others are treated as "Unimportant", when the importance of attributes is significantly lower than max_{z_score} . Variables which are considered "Unimportant" are permanently removed from the information system whereas the attributes

which are neither approved nor rejected are further marked as "Undetermined". Finally, the algorithm stops either when all features get confirmed or rejected or else it reaches a pre-defined limit of random forest runs (max_RF_runs). The time complexity of Boruta is approximately $O(P \cdot N)$, where P and N are the numbers of attributes and objects [30].

Algorithm 1. Feature Selection Using Boruta

input: Dataset D , Feature Set $F_S = \{f_1, f_2, \dots, f_t\}$

output: Selected feature set $F_{sel} = att[]$

```

1: Load dataset D
2:  $num_f \leftarrow att[].count$ 
3: for  $f = 1$  to  $num_f$  do
4:   Create shadow variable  $var_f^{shad}$ 
5:   Embed  $var_f^{shad}$  to the information system
6: end for
7:  $var_{shuf}^{shad} \leftarrow$  Randomly shuffle the  $var_f^{shad}$  across objects
8:  $z\_score[] =$  Call RF_Classifier( $train\_set, var_{shuf}^{shad}$ )
9: Initialize  $max = 0, r = 0$ ;
10: while  $r \leq max\_RF\_runs$  do
11:   for  $f = 1$  to  $num_f$  do
12:      $z\_score[] =$  Call RF_Classifier( $train\_set, att[f]$ )
13:      $max_{z\_score} \leftarrow z\_score[f]$ 
14:     if  $z\_score[f] > max_{z\_score}$  then
15:        $att[f] \rightarrow$  "Important"
16:     else if  $z\_score[f] < max_{z\_score}$  then
17:        $att[f] \rightarrow$  "Unimportant"
18:     else
19:       Perform two sided equality test
20:       ( $\forall att[f] \neq$  "Important") && ( $\forall att[f] \neq$  "Unimportant")
21:        $att[f] \rightarrow$  "Undetermined"
22:     end if
23:   end for
24: end while
25: return  $att[]$ ;

```

26: **function** **RF_Classifier**($train, feature$)

```

27: {
28:   Initialize  $T \rightarrow$  Number of trees in forest
29:   Set counter  $s = 0$ 
30:   for  $l=1$  to  $T$  do
31:     Select bootstrap sample of  $m$  variables from training dataset
32:      $\forall m$ , compute the importance  $var_{imp}$  by gini impurity criterion
33:     Build decision tree  $D_t$  for selected variables
34:     Initialize  $node_{min} \rightarrow$  minimum node size
35:     if  $s \leq node_{min}$  then
36:       Add node to  $D_t$ 
37:        $s+=1$ ;
38:     end if
39:      $node_{D_t} \leftarrow$  Number of nodes in  $D_t$ 
40:      $node_{imp} = \frac{\sum_{n=1}^{node_{D_t}} var_{imp}}{node_{D_t}}$ 
41:     Compute variance of nodes  $\sigma_{node}$  in  $D_t$ 
42:   end for
43:   for all  $T$  do
44:     Compute  $z\_score(D_t) = \frac{node_{imp}}{\sigma_{node}} \times \sqrt{node_{D_t}}$ 
45:      $z\_score[] \leftarrow z\_score(D_t)$ 
46:   end for
47:   return  $z\_score[]$ ;
48: }

```

3.2 Phase II: Locality Sensitivity Hashing (LSH) Based Support Vector Machine (SVM) Approach for Classification

SVM is a well known supervised learning approach that can be used for classification and regression analysis. In the formal context, SVM can be understood as a model that builds a hyperplane to separate binary classes by leveraging the benefits of a maximized margin criterion [31], [32].

Considering a linear separable training set, $T = \{x_i, y_i\}$ where $(i = 1, \dots, k)$. A hyperplane can be expressed as follows:

$$f(x) = \omega^T x_i + b = 0. \quad (1)$$

In the above equation, x_i are the set of training points (vectors) and $x_i \in R_d$ (for some dimension d) and b represents the real numbers. The above hyperplane aims to separate the training samples such that

$$\omega^T x_i + b \geq 0 \text{ for } y_i = +1 \quad (2)$$

$$\omega^T x_i + b < 0 \text{ for } y_i = -1. \quad (3)$$

A decent amount of separation can be achieved by the hyperplane which has the maximum distance from the closest training data point of the considered class. This is because higher margins lead to lower generalization errors of the underlying classifier. The best hyperplane that separates the data points is expressed using the following equation:

$$y_i f(x_i) \geq 1. \quad (4)$$

The point x_i which lie on the boundary of the hyperplane are referred to as support vectors, mathematically they are denoted using, $y_i f(x_i) = 1$, where $y_i \in \{-1, +1\}$ denotes the categories of x_i . Moreover, SVM finds an optimal solution for the following problem:

$$\begin{aligned} \min & \frac{1}{l} \omega^T \omega + C \sum_{i=1}^l \xi_i \\ \text{s.t. } & y_i (\omega^T \varphi(x_i) + b) \geq 1 - \xi_i, \quad \xi \geq 0, \forall i, \end{aligned} \quad (5)$$

where, $\omega \in R^d$ refers to the weight vector, $C \in R_+$ denotes the regularization constant and φ is the mapping function used to project the training data into the appropriate feature space using Boruta. This is done in order to allow the non-linear decision surfaces.

SVM is computationally efficient for learning good separating hyperplanes in a high dimensional feature space. It can be utilized both for linear and non-linear classification. In order to perform the classification, it utilizes the kernel functionalities which refer to the implicit mapping of given inputs to multi-dimensional feature space.

Definition 2 (Kernel Function). Suppose two vectors x_i and y_i have very high dimensional feature space. There is a mapping function $\varphi: \mathbf{R}^k \rightarrow \mathbf{R}^m$ that brings vectors of $\mathbf{R}^k$ to feature-space $\mathbf{R}^m$. Then the dot product of x_i and y_i in this space is $\varphi(x_i)^T \varphi(y_i)$. Hence, a kernel is defined as a function that corresponds to this dot product, i.e., $\mathcal{K}(x_i, y_i) = \varphi(x_i)^T \varphi(y_i)$ [33].

The role of kernel-function is to induce a feature space by implicitly mapping the training data into a higher dimensional

space where the data is linearly separable. However, due to the high dimensionality of kernel space, mapping of the original space S into the kernel space is usually not explicit. Further it also suffers from the huge space and time complexities involved due to the operations of the kernel matrix. Several research efforts have been made in order to overcome these challenges but the scalability issue of kernel functions is still a bottleneck for large-scale classification problems. Thus, a scalable Locality Sensitive Hashing based approach is proposed in this paper.

Definition 3 (LSH). For a training dataset T with input vectors (x_i, y_i) , LSH aims to project the training set T into low-dimensional Hamming space where each input data vector x_i is mapped to a b -bit vector, called a hash-key [34].

The basic idea of the LSH technique is to use the multiple hash functions to hash the data points. Thus there exists a high probability of collision for the points that are located close to each other whereas there is low collision probability for the points which are dissimilar. Here, the choice of hash functions $\mathcal{H}$ depends on similarity property:

$$Pr[h(x_i) = h(y_i)] = \text{sim}(x_i, y_i), \quad (6)$$

where $Pr[h(x_i) = h(y_i)]$ in the above equation denotes the expectation of identical hash bit over the possible hashing functions and $\text{sim}(x_i, y_i) \in [0, 1]$ represents the similarity function. In the case where data is kernelized, access to the data is only available through kernel function $\mathcal{K}$ where inputs are denoted as $\varphi(x_i)$. Now, to compute a hash function for a given problem, a random hyperplane in the kernel-induced feature space is required for reference which was given by Charikar in [35] as:

$$h_r(x_i) = \begin{cases} 1, & \text{if } \omega^T x_i \geq 0 \\ 0, & \text{otherwise} \end{cases}, \quad (7)$$

where r is a random hyperplane distributed according to the multivariate Gaussian $\mathcal{N}(0, 1)$ of the same dimensionality as x_i . Now hash function $h_\omega(x_i)$ uses the property of LSH as given below:

$$P[\text{sgn}(x_i^T \omega) = \text{sgn}(y_i^T \omega)] = 1 - \frac{1}{\pi} \cos^{-1} \left(\frac{x_i^T y_i}{|x_i| |y_i|} \right), \quad (8)$$

where, the random vector r is defined using central limit theorem as in [36]:

$$r = cov^{-1/2} \sqrt{k} \left(\frac{1}{k} \sum_{i \in T} x_i - \mu \right) \quad (9)$$

here, μ and cov are the mean and covariance of data vector x_i respectively. In all the previous works, hash functions have been designed for cases where similarity metric “sim” refers to the Mahalanobis metric. However, in this work, the similarity function of interest is the RBF kernel function which is given as [37]:

$$\text{sim}(x_i, y_i) = \frac{\mathcal{K}(x_i, y_i)}{(\sqrt{\mathcal{K}(x_i, x_i) \mathcal{K}(y_i, y_i)})}, \quad (10)$$

where $\mathcal{K}$ is the RBF kernel function given by the equation below:

$$\mathcal{K}(x_i, y_i) = \exp \frac{-|x_i - y_i|^2}{2\sigma^2}, \quad (11)$$

where $|x_i - y_i|^2$ denotes the squared euclidean distance between two vectors x_i and y_i whereas σ represents a free parameter. Now, LSH-SVM can be defined as:

Definition 4 (LSH-SVM). An LSH based SVM is defined as a technique for solving hyperplane queries efficiently in order to provide enhanced data separability to the classifier. By doing so, LSH fastens the prediction phase of the SVM decision function.

The key challenge in applying LSH to SVM is of constructing a vector ω from multivariate Gaussian $\mathcal{N}(0, 1)$ in such a manner that $\omega^T x_i$ can be computed through the RBF kernel function given in Eq. (11). In order to solve this problem, the proposed classification technique constructs a random vector ω as a weighted sum of input vectors x_i using Eq. (9). After the appropriate construction of ω , the random hyperplane hash function ($h_D(x_i)$) for ω is computed using the RBF kernel function as shown below:

$$h_D(x_i) = \sum_{\forall i} \omega \cdot \mathcal{K}(x_i, y_i). \quad (12)$$

Given a family of hash functions $\mathcal{F}$, $f: \mathbf{R}^k$, collision probability $\mathcal{P}$ between x_i and ω is defined using Eq. (8) as:

$$\mathcal{P}(x_i, \omega) = 1 - \frac{1}{\pi} \cos^{-1} \left(\frac{x_i \cdot \omega}{|x_i| |\omega|} \right). \quad (13)$$

Now, for the given SVM classifier in Eq. (1), the LSH based SVM classifier is defined as [38]:

$$\mathcal{C} = \begin{cases} h(x_i) = \text{sgn}(\xi_{\omega, b} - \mathcal{D}_H(h_D(x_i), h_D(\omega))) \\ \text{where } \xi_{\omega, b} = \frac{D}{\pi} \cos^{-1} \left(\frac{-b}{|\omega|} \right) \end{cases} \quad (14)$$

here, $h_D(x_i)$ and $h_D(\omega)$ are the hash codes for input vectors x_i and random variable ω respectively whereas $\mathcal{D}_H$ denotes the Hamming distance between them.

Notations		
1	Virtual machines (V)	index i
2	Server (S)	index j
3	Data centers (DC)	index k
4	Job (J)	index l

4 PHASE III: MULTI-OBJECTIVE OPTIMIZATION PROBLEM FOR JOB SCHEDULING AND VM PLACEMENT

The proposed job scheduling problem in geographically distributed DCs is formulated as a MOOP which caters to the following set of objectives i) maximal SLA adherence, ii) minimal energy consumption cost, iii) reduced carbon footprint rate, and iv) enhanced sustainability of DCs using RES. The considered problem involves the basic understanding of the following definitions:

Definition 5 (Server Power Consumption Modeling).

According to [3], a server's power consumption ($P_s^{U_{st}}$) increases linearly with its CPU utilization (U_{st}). It is mathematically formulated as follows:

$$P_s^{U_{st}} = P_{s, idle} + \left(P_{s, peak} - P_{s, idle} \right) \times U_{st}. \quad (15)$$

In the above equation, the variable $P_{s, idle}$ denotes the server's power consumption when idle and $P_{s, peak}$ represents the server's power utilization during peak CPU utilization. Further, U_{st} can be computed using the equation below:

$$U_{st} = \frac{(c_1 \times S^p) + (c_2 \times S^m)}{S^p + S^m} \times 100,$$

where, S^p and S^m refer to the server's processing and memory capacities respectively. On the other hand, the overall allocated portions of S^p and S^m are represented using the constants c_1 and c_2 .

Definition 6 (Power Usage Effectiveness (PUE)). This is an important metric used to quantify DC's power efficiency. It is often expressed as the total energy utilization of a DC to support its computing equipment ($\mathbf{P}_S^U$) relative to the other overheads like cooling and support systems ($\mathbf{P}_S^O$).

$$\text{PUE}(S_{jk}) = \frac{\mathbf{P}_S^T}{\mathbf{P}_S^U} = \frac{\mathbf{P}_S^O + \mathbf{P}_S^U}{\mathbf{P}_S^U}. \quad (16)$$

Definition 7 (SLA adherence). A SLA refers to the official commitment that entitles the service provider to maintain a definite level of quality, availability and responsibility; while providing the intended services. In this work, SLA met ($\text{SLA}(V_{ijk})$) is specified on the basis of resources (processing and memory capacity) that are allocated ($\mathbf{R}_{ijk}^{allo}$) to the VMs in comparison to the total number of required resources ($\mathbf{R}_{ijk}^{req}$), for successful job execution.

$$\begin{aligned} \text{SLA}(V_{ijk}) &= \left(\frac{\mathbf{R}_{ijk}^{allo}}{\mathbf{R}_{ijk}^{req}} \right) \times 100 \\ \text{s.t. } \mathbf{R}_{ijk}^{allo} &= (c_1 \times r_{ijkl}^p) + (c_2 \times r_{ijkl}^m) \\ \mathbf{R}_{ijk}^{req} &= r_{ijkl}^p + r_{ijkl}^m, \end{aligned} \quad (17)$$

where, r_{ijkl}^p and r_{ijkl}^m refer to the total processing and memory requirements for executing Job J on the i th VM of the j th server.

Definition 8 (Server energy utilization cost). The cumulative cost of energy utilization of a server ($\mathbf{C}$) is due to the cost of total energy utilization by the IT devices ($\mathbf{C}_s$) along with the cost of overhead energy to support the virtualization ($\mathbf{C}_o$). It is mathematically expressed using the following Equation [13].

$$\begin{aligned} \mathbf{C}(V_{ijk}) &= \left(\mathbf{C}_s(V_{ijk}) + \mathbf{C}_o(V_{ijk}) \right) \\ \text{s.t. } \mathbf{C}_s(V_{ijk}) &= \sum_{\tau \in \{S, W, NR\}} \mathbf{E}_s(V_{ijk}) \times \mathbf{C}_\tau^E \\ \mathbf{C}_o(V_{ijk}) &= \sum_{\tau \in \{S, W, NR\}} \mathbf{E}_o(V_{ijk}) \times \mathbf{C}_\tau^E \\ \mathbf{E}_s(V_{ijk}) &= \Delta P_S \times \mathbf{T}(V_{ijk}) \\ &= (P_{s, peak} - P_{s, idle}) (U_{t+1} - U_t) \times \mathbf{T}(V_{ijk}) \\ \mathbf{E}_o(V_{ijk}) &= \Delta P_o \times \mathbf{T}(V_{ijk}) \\ &= \Delta P_S \times (\text{PUE}(S) - 1) \times \mathbf{T}(V_{ijk}) \end{aligned} \quad (18)$$

In the above equations, $E_s(V_{ijk})$ represents the energy utilization of the VM to support IT equipment while $E_o(V_{ijk})$ denotes the overhead energy consumption. The parameter C_τ^E is the cost of energy (\$/MW) and τ represents the source of electricity generation, i.e., solar (S), wind (W) and non-RES (NR). Moreover, the change in power of the server (due to deployment of VM) from time t to $t + 1$ is denoted using ΔP_s and the variable $T(V_{ijk})$ refers to the holding time of the VM.

Definition 9 (Carbon foot print rate). *The carbon intensity (R) varies significantly with source of electricity generation. Here, the variables R_E^S , R_E^W , and R_E^{NR} have been used to represent carbon intensity of the energy source – solar, wind and non-RES. The metric carbon intensity is expressed in tons per mega watt hour electricity used (Tons/MWh). Higher values of carbon intensity metric denote higher emission of carbon dioxide in environment which is represented as follows [13]:*

$$R(V_{ijk}) = \sum_{\tau \in \{S, W, NR\}} \left(E_{s,\tau} + E_{o,\tau} \right) \times R_\tau^E. \quad (19)$$

Definition 10 (VM mapping variable). *A binary decision variable, i.e., X_{ijk} has been used to represent the VM mapping. It is represented using the following equation:*

$$X_{ijk} = \begin{cases} 1 : & \text{If } i\text{th VM is mapped on the } j\text{th server} \\ 0 : & \text{Otherwise} \end{cases},$$

where, the variable X_{ijk} is assigned the value of 1 if the i th VM is mapped on the j th server of the k th DC. Otherwise, X_{ijk} is set to the value of 0.

Definition 11 (Job scheduling variable). *A binary decision variable for job scheduling (Y_{ijkl}) is represented using the equation below:*

$$Y_{ijkl} = \begin{cases} 1 : & \text{If } l\text{th job is scheduled on } i\text{th VM} \\ 0 : & \text{Otherwise} \end{cases}.$$

As per the above equation, the value of Y_{ijkl} is set to 1 if the l th job is scheduled on i th VM of the k th DC, else it's set to 0.

4.1 List of Objectives

The proposed scheme has the following set of objectives:

4.1.1 Maximal SLA Adherence and QoS Guarantee

One of the important concerns of the proposed scheme is to ensure maximal SLA adherence; while scheduling the bunch of jobs on the available geographically distributed servers using virtualization. It is mathematically expressed as follows using the objective function $F_1(X_{ijk}, Y_{ijkl})$.

$$F_1(X_{ijk}, Y_{ijkl}) = \sum_{i \in V} \sum_{j \in S} \sum_{k \in DC} \sum_{l \in J} \left(SLA(V_{ijk}) \right) \times X_{ijk} \times Y_{ijkl}. \quad (20)$$

4.1.2 Minimal Energy Cost

This objective function ($F_2(X_{ijk}, Y_{ijkl})$) deals with minimizing the total energy cost of the DCs while achieving desirable job scheduling and VM mapping. It is achieved in

accordance with the following equation:

$$F_2(X_{ijk}, Y_{ijkl}) = \sum_{i \in V} \sum_{j \in S} \sum_{k \in DC} \sum_{l \in J} \left(C(V_{ijk}) \right) \times X_{ijk} \times Y_{ijkl}. \quad (21)$$

4.1.3 Minimal Carbon Footprint Rate

The scope of the present objective function ($F_3(X_{ijk}, Y_{ijkl})$) is to ensure the maximal use of RES to reduce the carbon footprint rate. Its mathematical formulation is expressed as follows [13]:

$$F_3(X_{ijk}, Y_{ijkl}) = \sum_{i \in V} \sum_{j \in S} \sum_{k \in DC} \sum_{l \in J} \left(R(V_{ijk}) \right) \times X_{ijk} \times Y_{ijkl}. \quad (22)$$

4.1.4 Maximal RES Sustainability

The primary goal of the stated objective function ($F_4(X_{ijk}, Y_{ijkl})$) is to sustain a DC's operational activities using RES to the maximum extent. This in turn reduces the dependence of DCs on power grids to meet their day-to-day operational expenses. This is ensured using the equation below:

$$F_4(X_{ijk}, Y_{ijkl}) = E^{tot} - E^{ren}, \quad (23)$$

wherein, E^{tot} represents the total energy utilization of all the DCs and variable E^{ren} refers to the total available from RES (across all DCs) to support a DC's operational and computational activities. The values of E^{tot} and E^{ren} are computed as follows:

$$E^{tot} = \sum_{j \in S} \sum_{k \in DC} E_{jk}^{tot}; \quad E^{ren} = \sum_{k \in DC} E_k^{ren}.$$

In the above equations, E_{ij}^{tot} and E_i^{ren} represent the total energy utilization of the j th server and the total energy generated from all the RES across k th DC respectively.

4.2 List of Constraints

In view of the above mentioned objective functions, the following set of constraints have been defined. Their detailed mathematical representation is illustrated as follows:

$$C1 : \sum_{i \in V} \sum_{j \in S} \sum_{k \in DC} r_{ijk}^p \times X_{ijk} \leq S^p$$

$$C2 : \sum_{i \in V} \sum_{j \in S} \sum_{k \in DC} r_{ijk}^m \times X_{ijk} \leq S^m$$

$$C3 : X_{ijk} \in \{0, 1\}; \forall i, j, k$$

$$C4 : Y_{ijkl} \in \{0, 1\}; \forall i, j, k, l.$$

In the above equations, constraints C1 and C2 impose an upper bound on the number of resources that can be allocated by a server to its underlying VMs. These resources are processing (S^p) and memory units (S^m). Also, C3 and C4 impose an integrality restriction on X_{ijk} and Y_{ijkl} decision variables.

4.3 Multi-Objective Optimization Problem (MOOP)

It is evident from the above discussion that the job scheduling problem on cloud DCs under consideration is a special case of energy-aware and SLA-driven strategy, with

emphasis on sustainability of DCs using RES. This MOOP can be formulated using the below mentioned objective function ($F(\mathbf{X}_{ijk}, \mathbf{Y}_{ijkl})$) in the multi-constraint setup.

$$\begin{aligned} F(\mathbf{X}_{ijk}, \mathbf{Y}_{ijkl}) &= \min f(-F_1, F_2, F_3, F_4) \\ \text{s.t. Constraints C1} - \text{C6 hold.} \end{aligned} \quad (24)$$

However, the above problem involves complex interactions with numerous subsystems such as job scheduler, energy dispatcher and decision maker. Hence, an efficient algorithm based on a greedy heuristic is presented in the next section for efficient job scheduling and VM placement in polynomial time. In the considered setup, the greedy approach has been recognized over the widely available meta-heuristic approaches, mainly due to its ability to converge faster, and produce local optimal solutions (which often coincide with the optimal global solution). In contrast, the present meta-heuristics tend to diligently explore the search space for tracking the near optimal solution. However, this comes at the cost of reduced convergence speed and non-determinism. Additionally, meta-heuristics are generally not problem-specific and tend to give unsatisfactory results. Further, prolonged time for convergence to the nearoptimal solution may prove to be an ineffective approach with the Cloud's on-demand service for job scheduling and VM placement.

4.4 Enhanced Heuristic Approach for Job Scheduling and VM Placement

The crux of the designed algorithm lies in leveraging the concept of data fields and priority queues to achieve near optimal results. The detailed technical description is described below.

4.4.1 Potential Utility Value of Servers

In the scenario under consideration of geo-distributed DCs, the selection of the optimal servers for VM placement and job scheduling is a challenging task. Hence, a greedy strategy has been used to select the servers on the basis of their current resource and energy utilization. The overall idea is to schedule jobs on those servers first which have high utilization levels in comparison to the other set of servers. This in turn helps to minimize overall energy consumption of DCs by utilizing the servers to their best possible capacity. Apparently, the availability of higher RES at the DCs is also essential in considering the optimal server for job scheduling. This in turn helps to attain our last objective for self-sustainable DCs. Hence, in order to achieve maximal RES sustainability, potential utility (PU_{jk}) values of all the servers are computed keeping in mind the above considerations. So, PU_{jk} is expressed as follows:

$$\text{PU}_{jk} = \sum_{i \in V} \left(\frac{r_{ijk}^p}{r_{jk}^p} \times \frac{r_{ijk}^m}{r_{jk}^m} \times \frac{E_{ijk}}{E_{jk}^{\max}} \right) \times e^{-(R(V_{ijk}))^2}. \quad (25)$$

In the above equation, r_{jk}^p and r_{jk}^m are the total processing and memory capacities of the j th server. On the other hand, E_{ijk} denotes the energy utilization for executing the i th VM on j th server at k th DC. Similarly, the maximum energy utilization of the j th server is denoted using $E_{jk}^{\max}$. Here, higher

PU_{jk} values indicate servers with higher CPU utilization levels and greater RES.

Algorithm 2. Proposed Job Scheduling and VM Placement Algorithm

Input: $\mathcal{J}, \mathcal{V}, \mathcal{S}$ and $\mathcal{DC}$.

Output: $\mathbf{X}_{ijk}$ and $\mathbf{Y}_{ijkl}$

```

1: procedure FunctionVM-PLACEMENT-&-JOB-SCHEDULING
2:   Step 1: Feature Selection
3:   {Selected features}  $\leftarrow$  Call Algorithm 1
4:   Step 2: Classification
5:    $\{\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3\} \leftarrow$  Using Eq. (14)
6:   Step 3: VM Placement & Job Scheduling
7:   Compute  $\text{PU}_{jk}$  using Eq. (25);  $\forall j, k$ 
8:    $\mathbf{S}^{\text{PU}} \leftarrow$  Sort  $\mathbf{S}$  according to decreasing value of  $\text{PU}_{jk}$ 
9:   while  $|\mathbf{Q}_1| \neq \text{null} \ \&\& \ |\mathbf{Q}_2| \neq \text{null} \ \&\& \ |\mathbf{Q}_3| \neq \text{null}$  do
10:     $y \leftarrow \text{first}(\mathbf{S}^{\text{PU}})$ 
11:     $z \leftarrow \text{pop}(\mathbf{Q}_1)$ 
12:     $x \leftarrow \mathcal{VM}(z)$ 
13:    if  $(r_{xz}^p \leq \text{Avail}(S_y^p)) \ \&\& \ (r_{xz}^m \leq \text{Avail}(S_y^m))$  then
14:      Set  $\mathbf{X}_{xy} \leftarrow 1$ 
15:      Set  $\mathbf{Y}_{xyz} \leftarrow 1$ 
16:    else
17:       $\hat{y} \leftarrow y$ 
18:       $y \leftarrow \text{last}(\mathbf{S}^{\text{PU}})$ 
19:      if  $(r_{xz}^p \leq \text{Avail}(S_y^p)) \ \&\& \ (r_{xz}^m \leq \text{Avail}(S_y^m))$  then
20:        Set  $\mathbf{X}_{xy} \leftarrow 1$ 
21:        Set  $\mathbf{Y}_{xyz} \leftarrow 1$ 
22:      else
23:        Job is pushed into the waiting queue
24:      end if
25:       $\hat{y} \leftarrow y$ 
26:       $y \leftarrow \hat{y}$ 
27:       $z \leftarrow \text{pop}(\mathbf{Q}_2)$ 
28:       $x \leftarrow \mathcal{VM}(z)$ 
29:      if  $(r_{xz}^p \leq \text{Avail}(S_y^p)) \ \&\& \ (r_{xz}^m \leq \text{Avail}(S_y^m))$  then
30:        Set  $\mathbf{X}_{xy} \leftarrow 1$ 
31:        Set  $\mathbf{Y}_{xyz} \leftarrow 1$ 
32:      else
33:        Repeat Steps 17-26
34:       $z \leftarrow \text{pop}(\mathbf{Q}_3)$ 
35:       $x \leftarrow \mathcal{VM}(z)$ 
36:      if  $(r_{xz}^p \leq \text{Avail}(S_y^p)) \ \&\& \ (r_{xz}^m \leq \text{Avail}(S_y^m))$  then
37:        Set  $\mathbf{X}_{xy} \leftarrow 1$ 
38:        Set  $\mathbf{Y}_{xyz} \leftarrow 1$ 
39:      else
40:        Repeat Steps 17-24
41:      end if
42:    end if
43:  end while
44:  end procedure
45:  Return  $\mathbf{X}_{ijk}$  and  $\mathbf{Y}_{ijkl}$ 

```

4.4.2 Designed Algorithm

The designed algorithm works as per the following sequence based on Algorithm 2. Its overall working can be segregated broadly into three steps. In the first step, the relevant set of features is identified from the given dataset using

Algorithm 1 (Lines 2-3). In the next step, classification is carried out to prioritize the jobs on the basis of features selected in the previous step. Accordingly, submitted jobs are prioritized into three categories, and are then pushed to different queues (Q_1, Q_2, Q_3) in decreasing order of priority (Lines 4-5). Finally, the last step is to schedule the jobs on VMs while achieving near optimal results as per the MOOP defined in Eq. (24). Hence, in the first phase, PU_{jk} values of all the servers are computed. Finally, the servers are sorted according to the decreasing values of PU_{jk} in the list S^{PU} (Lines 6-8). Next, job scheduling is initialized with the Q_1 , wherein available resources of the y th server ($Avail(S_y^p), Avail(S_y^m)$) are compared against the required resources for z th job ((r_{xz}^p, r_{xz}^m)). In case, the resources are found to be insufficient, then the y th server is selected either for scheduling the topmost job from Q_2 or Q_3 . Further, for scheduling the z th job (from Q_1), the last server of the list S^{PU} (i.e., the least utilized server) is chosen (Lines 8-46). This process is repeated until all the jobs are scheduled.

• *Complexity Analysis*: The overall complexity of the proposed algorithm is dependent upon the complexities to carry out feature selection (using *Boruta*), workload classification (using *LSH-based SVM*), and job scheduling (using the proposed heuristic approach). The overall complexity of *Boruta*-based feature selection mechanism is equivalent to $O(P \cdot N)$, where P and N are the numbers of attributes and objects. On the other hand, the complexity of the employed classification mechanism is equivalent to $O(n_{sv} \cdot d)$, where n_{sv} denotes number of support vectors and d refers to number of input dimensionality. Finally, the complexity of the proposed job scheduling and VM placement procedure is dependent upon of the length of queues Q_1 , Q_2 , and Q_3 , respectively. Thus, the overall complexity of the proposed Algorithm 2 is expressed as follows:

$$\begin{aligned} \text{Complexity} &= O(P \cdot N) + O(n_{sv} \cdot d) + O(|Q_1| + |Q_2| + |Q_3|) \\ &= O(P \cdot N) + O(n_{sv} \cdot d) + O(M). \end{aligned} \quad (26)$$

The computed complexity is the overall overhead of the proposed algorithm for attaining efficient job scheduling in sustainable cloud DCs.

5 RESULTS AND DISCUSSION

This section presents the results obtained using the proposed scheme and its relative comparison with the current state-of-the-art schemes. For the purpose of evaluation of the proposed scheme three different case studies have been taken into account. The first case study compares the performance of the proposed scheme with the standard First Come First Serve (FCFS) approach. In contrast, the second and third case study present the evaluation of the proposed scheme against the existing state-of-the-art techniques. Its detailed explanation is presented as follows along with the datasets employed.

5.1 Case Study I

In this case study, real-time data traces have been acquired from Google Cluster [39] for simulation purposes. In the setup under consideration, a total of four heterogeneous DCs are assumed to be dispersed across different

TABLE 2
Accuracy of SVM Across Different Kernels

Kernel Function	8 Features					
Linear	100	100	100	100	66.54	87.6
Polynomial	100	100	100	100	67.33	88.62
RBF	100	100	100	100	70.32	90.43

geographical locations. Each DC is equipped with a total of 24 racks of 15 servers. Apart from this, two DCs are supported by PV panels while the other two are sustained using energy generated from wind. Moreover, to evaluate the performance of the proposed scheme, placement of jobs on different servers using FCFS approach is referred as “Existing Scheme”. The detailed performance evaluation results are presented in the next section.

5.1.1 Performance Evaluation

In the initial phase, the proposed scheme primarily concentrates on feature selection and classification approach to derive the relative weights of significant features. This is done using the RBF kernel which is found to perform better than linear and polynomial kernels as evident from the results drawn in Table 2.

This is done in order to classify the incoming workloads into three different classes; which in turn are associated with definite priorities. Subsequently, they are moved to the available set of queues $\{Q_1, Q_2, Q_3\}$ and the related results are shown in Fig. 2a. For instance, at 0500 hours a total of 37, 362 and 541 jobs were assigned to Q_1, Q_2 , and Q_3 respectively. These jobs were scheduled across different servers dispersed across geographically distributed DCs. This was particularly achieved in accordance with the availability of solar and wind energy at different locations. As evident from Fig. 2b, solar energy of 14 and 0.43 kW was available at DC_1 and DC_2 respectively at 0500 hours. At the same time-stamp, energy generated from wind was almost equivalent to 11 and 10.1 kW at DC_3 and DC_4 respectively.

Using the above mentioned statistics, Algorithm 2 achieves the desirable trade-off between the competing objectives as defined in Section 4.4. Hence, the incoming jobs were provisioned on different VMs which in turn were placed on the available array of servers. The related results have been highlighted using Fig. 2c across a 24 hour time interval. It is clear from the figure that the proposed scheme utilizes comparatively fewer servers for effective job scheduling relative to the existing scheme. For instance, at 0500 hours, a total of 347 servers were held in active mode to cater for the available workload by the proposed scheme, which was considerably less in comparison the existing scheme (472). Subsequently, this results in higher CPU utilization of servers provisioned using the proposed scheme in comparison to the existing scheme. These results are shown in Fig. 2d, which depicts the average utilization levels of servers across a time interval.

Based on the utilization levels of the servers, their total energy consumption is computed using Eq. (15). The obtained results have been illustrated using Fig. 2e. For instance, a total of 84 kWh energy was consumed by the

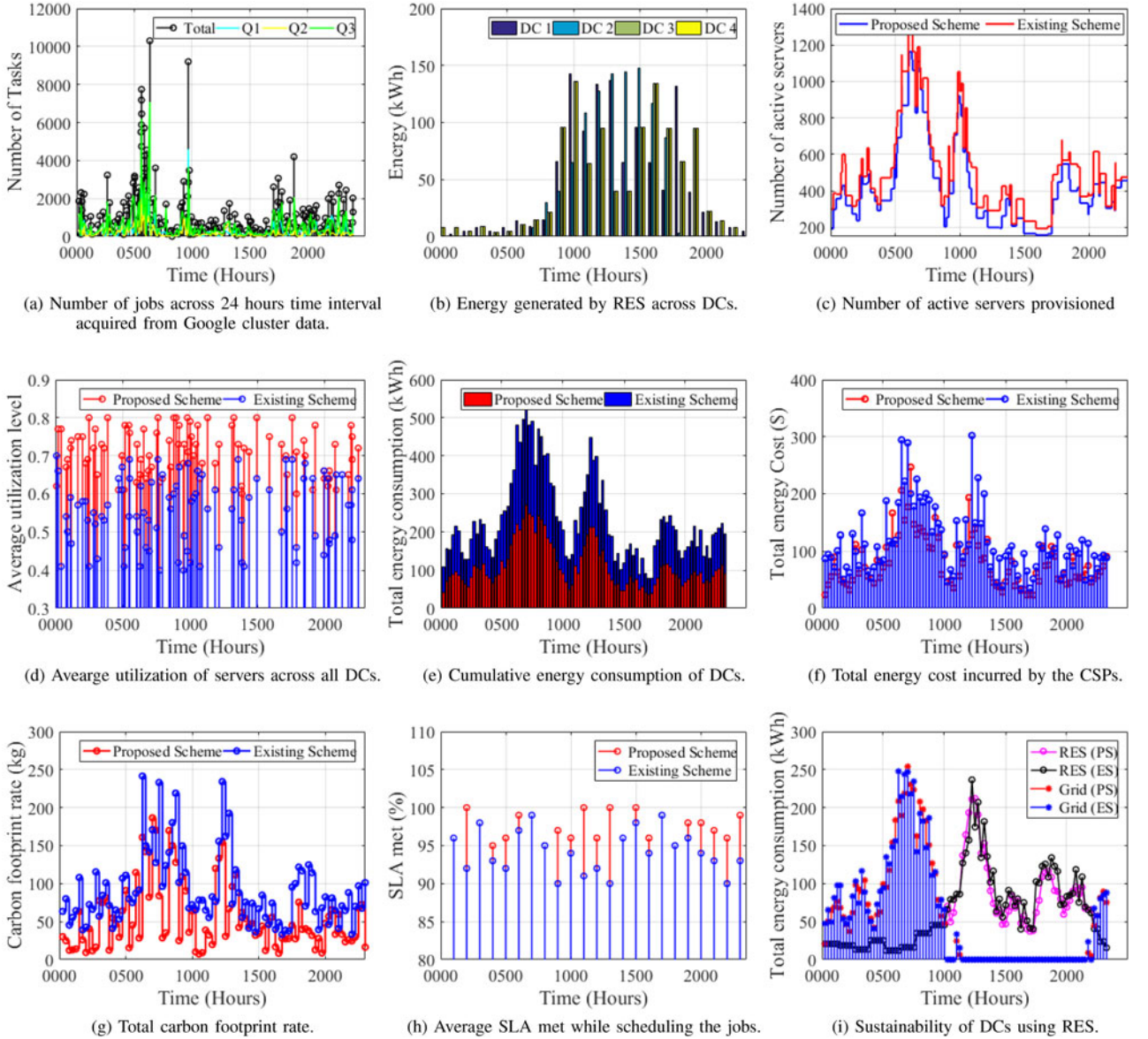


Fig. 2. Case Study I: Performance evaluation of the proposed scheme against the existing scheme.

bunch of active servers at 0500 hours using the proposed scheme. However, due to the large range of active servers provisioned using the existing scheme, a total 98 kWh energy was consumed by the servers dispersed across different DCs. On an average, the proposed scheme leads to an overall decrease of 38 percent energy consumption relative to the existing scheme. Subsequently, the overall energy expenditure across different DCs was also evaluated as one of the objectives of the proposed scheme. The obtained results have been depicted using Fig. 2f, wherein the proposed scheme results in overall decrease of 29.5 percent in energy prices relative to the existing scheme.

The use of electricity to support DC operations is also accompanied by carbon emission. Such a case occurs when the DC operations are met using electricity generated from non-RES; the higher the usage, the higher is the CFR as evident from Fig. 2g. Since the proposed scheme aims to achieve self-sustainability of DCs, reduction in CFR is one of the key objectives, and the proposed scheme achieves an overall reduction of 39.4 percent relative to the existing scheme.

Another significant objective of the proposed scheme was to attain maximal SLA adherence. The proposed scheme performed reasonably well in this case also, wherein an overall improvement of 2.9 percent has been witnessed (Fig. 2h). Moreover, the proposed scheme also consumed reasonably more amount of RES energy in comparison with the existing scheme. It is evident from the results depicted using Fig. 2i.

5.2 Case Study II

In this case study, the extended version of the CloudSim toolkit [40] has been chosen for simulation and performance evaluation of the proposed scheme against an existing scheme proposed in [13]. The simulation setup is the same as considered in [13].

5.2.1 Performance Evaluation

In the initial phase, the proposed scheme primarily concentrates on feature selection and classification approach to derive the relative weights of significant features. This is

TABLE 3
Accuracy of SVM across Different Kernels

Kernel Function	8 Features					
Linear	100	100	100	100	67.36	84.7
Polynomial	100	100	100	100	65.37	80.13
RBF	100	100	100	100	100	92.23

done using the RBF kernel which is found to perform better than linear and polynomial kernels as evident from Table 3.

The relative comparison of the proposed scheme against the existing scheme for the next phase is shown in Fig. 3. The experimental evaluations show an average improvement of 31 percent in energy utilization, 28.1 percent in energy cost, and 36.8 percent in CFR, with a slight degradation in SLA assurance (about 1.4 percent) compared with the existing scheme.

5.3 Case Study III

The cloud computing infrastructure essentially provides an infinite bunch of resources. Hence it is thereby necessary to evaluate any cloud-centric scheme on a large-scale virtualized DC. Nevertheless, repeatable experiments on a real-time setup are a highly complicated process that involves a lot of capital investment along with time [41]. Thus, to perform near to real-time experiments, and validate the efficacy of the proposed job scheduling approach relative to the existing state-of-the-art schemes, the prototype of the proposed approach has been simulated on Java-based simulator named CloudSim [40]. It enables energy-aware simulation in a cloud-based ecosystem with the provision to alter the incoming workloads, and configuration of cloud DCs, VMs, and hosts. The considered setup, following parameter settings, have been taken into account:

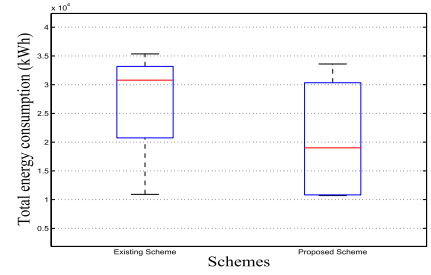
5.3.1 Parameter Settings

A cloud DC comprising of almost 1000 heterogeneous physical servers have been taken into account. Out of which half of the nodes are IBM System x3550 M3 (Intel Xeon X5675) while rest belong to IBM System x3550 M3 (Intel Xeon X5670) series. All the servers are enabled with a network bandwidth of 1 GB/sec. Further, the VMs closely resemble Amazon EC2 instances with the provision of single core and variable RAM allocations (ranging from 512 MB to 4 GB). Further, for effective performance evaluation, it is also necessary to perform the experiments on real-time data traces. In our assessments, we have chosen PlanetLab dataset for April 2011. These traces correspond to the marks of thousands of VMs mapped across servers from different geographical regions.

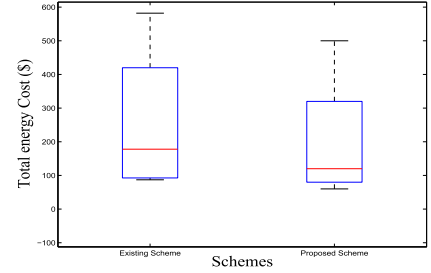
5.3.2 Existing State-of-the-Art

To evaluate the performance of the proposed job scheduling algorithm, four different existing algorithms have been chosen. The existing algorithms have been proposed in the works of [22], [23], [26], [27]. Their detailed description has already been provided in Section 1 with the detailed comparison of the characteristics supported by the algorithms. For the sake of simplicity, these algorithms are referred to as existing schemes (ES) throughout the manuscript, i.e., ES1, ES2, ES3, and ES4 respectively.

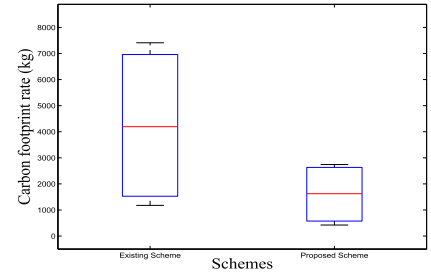
Authorized licensed use limited to: PES University Bengaluru. Downloaded on September 17, 2025 at 07:13:17 UTC from IEEE Xplore. Restrictions apply.



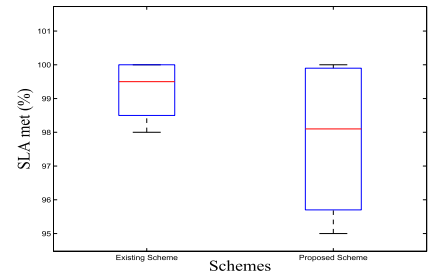
(a) Cumulative energy consumption of DCs.



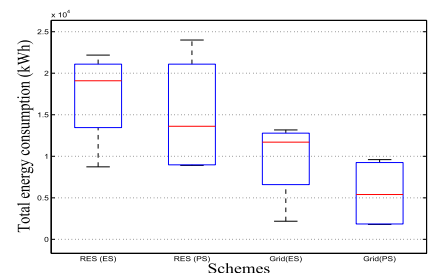
(b) Total energy cost incurred by the CSPs.



(c) Total carbon footprint rate.



(d) Average SLA met while scheduling the jobs.

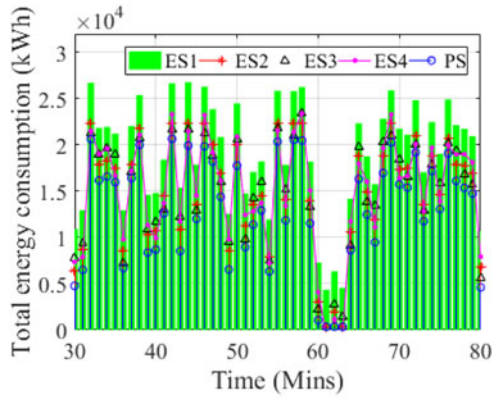


(e) Sustainability of DCs using RES.

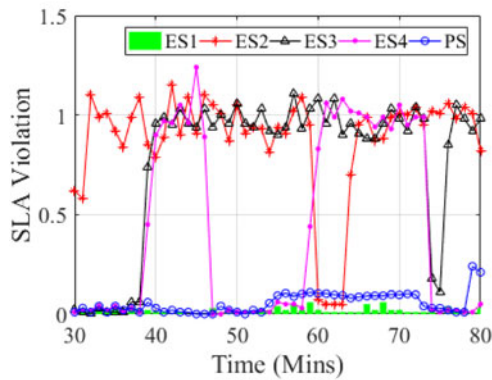
Fig. 3. Case Study II: Performance evaluation of the proposed scheme against the existing scheme.

5.3.3 Results Obtained

In order to evaluate the ES with the proposed schemes (PS), two different evaluation metrics have been considered namely-energy consumption and SLA violation. The corresponding results have been depicted using Fig. 4. For



(a) Cumulative energy consumption of cloud DC.



(b) SLA violation comparison.

Fig. 4. Case Study III: Performance evaluation of the proposed job scheduling approach against the existing schemes.

instance, the cumulative energy consumption of cloud DC is found to be highest in ES1 as shown in Fig. 4a. This is because ES1 does not give any emphasis towards energy minimization and targets to reduce the SLA violations during job scheduling. On the contrary, minimum energy consumption is achieved by the proposed algorithm followed by an almost comparable performance by ES2, ES3, and ES4. The SLA violations are found to be least using ES1 as indicative from the results drawn in Fig. 4b followed by the proposed algorithm.

In addition to the above comparison, we also compare the performance of the designed scheduler based on a

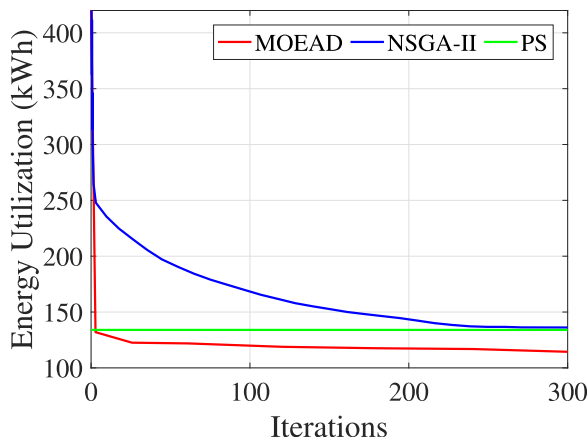


Fig. 5. Convergence evaluation.

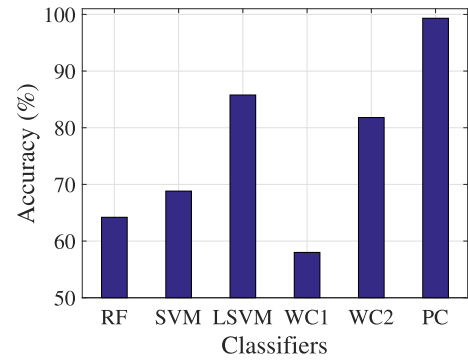


Fig. 6. Accuracy evaluation.

greedy strategy with the existing meta-heuristics based on NSGA-II and MOEA/D. The comparison has been done using the convergence speed evaluation as depicted in Fig. 5. As evident from the figure, the proposed algorithm may not achieve the most optimal solution but converges to the optimal local solution quickly. For instance, in the considered scenario, the sub-optimal solution obtained by the proposed scheme is far from the global optimal solution by almost 14%. In addition to this, the proposed scheme is based on greedy strategy to deduce the sub-optimal results, however, it does not undergo multiple iterations like NSGA-II and MOEA/D to deduce the optimal solution. Thus, on the basis of number of iterations, it would be correct to say that the proposed scheme attains its sub-optimal result in the first iteration itself; while the other existing approaches may take more than 200 iterations to attain their respective optimal results (as indicative from Fig. 5).

We also compared the performance of the proposed workload classification scheme with the traditional SVM classifier and random forest (RF) classifier. Moreover, to showcase the impact of feature selection on the proposed scheme, it has been validated against two simultaneous scenarios, LSH+SVM without Boruta (referred as *LSVM*), and LSH+SVM with Boruta (this is the proposed workload classification approach, referred as *PC*). Along with this, two existing schemes [42], [43] are considered during this assessment, referred to as WC1 and WC2, respectively. Here, the former considers the classification of the incoming workloads into three categories, namely, permanent, tiny, and normal. The classification mechanism predominantly considers the lifetime of the incoming workloads. On the contrary, the latter, WC2, adopts a mechanism based on k-means clustering for workload classification. The results for

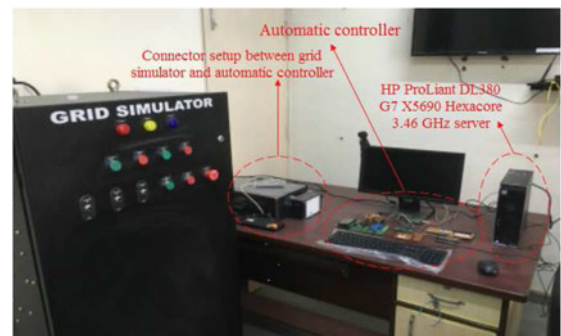


Fig. 7. Prototype deployment.

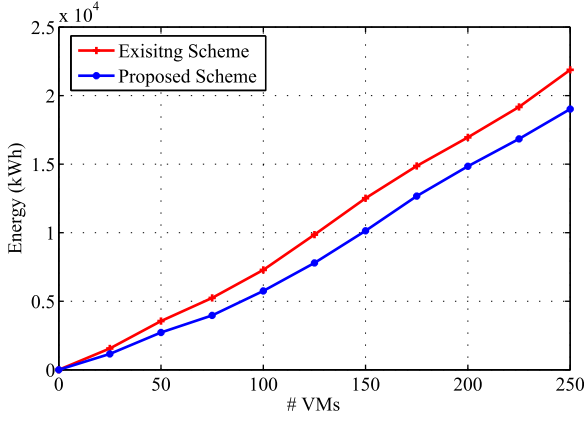


Fig. 8. Cumulative energy consumption.

the same has been demonstrated in Fig. 6. As expected, the accuracy of the proposed classification scheme using Boruta algorithm is quite high as compared to others.

5.4 Case Study IV: Prototype Deployment

The proposed scheme has also been evaluated over a real prototype model deployed in the smart grid testbed at Thapar Institute of Engineering and Technology, Patiala, Punjab, India as shown in Fig. 7. This prototype includes a STM8S micro controller, a RF transceiver, MAX 232 IC, and various types of relays wherein the real time data traces are converted into jobs which are scheduled to 5 HP ProLiant DL380 G7 X5690 Hexacore 3.46 GHz servers with a flash memory of 8-16 GB. More details about the testbed can be found at [44], [45].

The comparison of scalability of proposed scheme with respect to an increase in the number of VMs is evaluated. The amount of energy consumed is lesser as compared to the existing scheme as given in Fig. 8. On the similar trends, Fig. 9 represents the comparison between the SLA violations and the number of VMs in the server. The figure shows the superiority of the scheme as compared to the existing scheme even with an increase in the number of VMs.

6 CONCLUSION

In order to handle millions of incoming jobs, cloud DC market has shown significant growth. This in turn is accompanied with mega watts of power consumption and has led to substantial increase in the emission of carbon. Hence, shifting the paradigm from grid dependability to self-sustainability particularly for cloud DCs using renewable energy sources is an important research initiative. So, in this paper, a MOOP is formulated to encourage sustainability of geo-distributed cloud DCs. The designed approach operates in two phases. In the first phase, the incoming job requests are classified using a LSH-based SVM approach; which is preceded by selection of relevant features using Boruta. In the next phase, a MOOP is formulated on the basis of following competing parameters, i.e., SLA, energy cost, CFR, and availability of RES. The proposed scheme is evaluated using extensive simulations on real-time data traces acquired from Google Cluster for different case studies. The results obtained indicate convincing trade-off with respect to different parameters

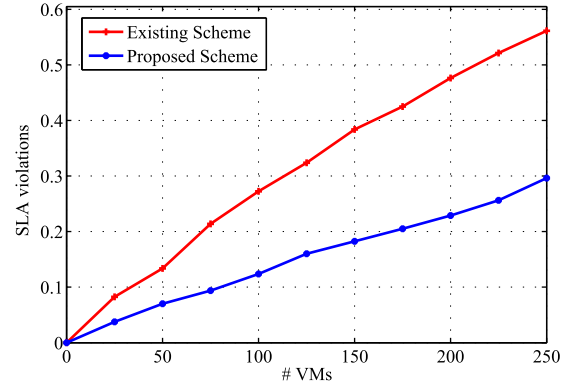


Fig. 9. SLA violations.

such as energy minimization (38 percent), increased SLA adherence (2.9 percent), energy cost reduction (29.5 percent), and CFR minimization (39.4 percent). Moreover, three additional case studies were performed to cross validate the effectiveness of the proposed scheme.

In future, the proposed scheme can be extended to purely green DCs powered with RES. The underlying job scheduling and VM placement problem would become more challenging due to the intermittent nature of RES.

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Kuljeet Kaur (S'13, M'18) received the PhD degree from the Thapar Institute of Engineering and Technology, Punjab, India in 2018. She is currently working as an NSERC post-doctoral research fellow with École de technologie supérieure, Université du Québec, Montréal, Canada. She has many research contributions in the areas of cloud computing, energy efficiency, smart grid, frequency support, and vehicle-to-grid. Some of her research findings are published in top cited journals such as the *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Smart Grid*, *IEEE Transactions on Cloud Computing*, *IEEE Transactions on Vehicular Technology*, the *IEEE Transactions on Network and Service*, the *IEEE Systems Journal*, *IEEE Transactions on Sustainable Computing*, the *IEEE Communications Magazine*, *IEEE Wireless Communications*, the *IEEE Network*, *IEEE PS*, *Future Generation Computing Systems*, the *Journal of Parallel and Distributed Computing*, and the *Peer-to-Peer Networking and Applications*, etc., and various International conferences including IEEE Globecom, IEEE ICC, IEEE PES GM, IEEE WCNC, IEEE Infocom Workshops, ACM Mobicom Workshops, ACM MobiHoc Workshops, etc. During her PhD, she received two prestigious fellowships including INSPIRE fellowship from Department of Science & Technology, India (in 2015) and research scholarship from Tata Consultancy Services (TCS) (from 2016–2018). She also received the IEEE ICC best paper award in 2018 at Kansas City, USA. She serves as an Associate Editor of Wiley's Security and Privacy (SPY) and Springer's Journal of Information Processing Systems (JIPS). She is a website co-chair of N2Women community. She has also been a TPC Co-chair for IEEE Infocom and ACM MobiCom workshop on DroneCom. She is a member of IEEE, IEEE Communications Society, IEEE Computer, IEEE Women in Engineering, IEEE Software Defined Networks Community, IEEE Smart Grid Community, ACM and IAENG.



Sahil Garg (S'15, M'18) received the PhD degree from the Thapar Institute of Engineering and Technology, Patiala, India, in 2018. He is currently working as a post-doctoral research fellow with École de technologie supérieure, Université du Québec, Montréal, Canada. His research interests include the area of machine learning, big data analytics, security & privacy, Internet of Things, and cloud computing. Some of his research findings have been published in top cited journals such as the

IEEE Transactions on Industrial Informatics, *IEEE Transactions on Multimedia*, *IEEE Transactions on Cloud Computing*, *IEEE Transactions on Network and Service*, *IEEE Transactions on Vehicular Technology*, the *IEEE Systems Journal*, the *IEEE Internet of Things Journal*, the *IEEE Transactions on Sustainable Computing*, *IEEE Communications Magazine*, the *IEEE Network*, the *IEEE Wireless Communications*, *IEEE Consumer Electronics Magazine*, *Future Generation Computing Systems*, *Journal of Parallel and Distributed Computing*, and the *Information Sciences* including various International conferences of repute such as IEEE Globecom, IEEE ICC, IEEE WCNC, IEEE CCNC, IEEE VTC, IEEE Infocom Workshops, ACM MobiCom Workshops, ACM MobiHoc Workshops, etc. He was the recipient of prestigious Visvesvaraya PhD fellowship from the Ministry of Electronics & Information Technology under Government of India (2016–2018). For his research, he has also been awarded the IEEE ICC best paper award in 2018 at Kansas City, USA. He serves as the Managing Editor of Springer's Human-centric Computing and Information Sciences (HCIS) and Associate Editor of Wiley's International Journal of Communication Systems (IJCS), and is serving as the Workshops and Symposia Officer for the IEEE ComSoc Emerging Technology Initiative on Aerial Communications. He has been a lead guest editor for a number of special issues in prestigious journals and magazines such as IEEE TII, IEEE T-ITS, IEEE IoT Journal, IEEE Network, and Elsevier's FGCS. He serves as the workshop chair/publicity co-chair for several International conferences including IEEE Infocom, ACM MobiCom, IEEE Globecom and IEEE ICC. He is a member of IEEE, IEEE Communications Society, IEEE Computer Society, IEEE Industrial Electronics Society, IEEE Software Defined Networks Community, IEEE Smart Grid Community, ACM, and IAENG.



Gagangeet Singh Aujla (S'15, M'18, M'19) received the BTech and MTech degrees from Punjab Technical University, Punjab, India, in 2003 and 2013, respectively, and the PhD degree from Thapar Institute of Engineering & Technology, India, in 2018, all in computer science and engineering. He is currently working as a post-doctorate research associate with Newcastle University, United Kingdom and an associate professor in computer science and engineering, Chandigarh University, India. He received 2018 IEEE TCSC

Outstanding PhD dissertation Award. Some of his research findings are published in top cited journals such as the *IEEE Transactions on Knowledge & Data Engineering*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Services Computing*, the *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Cloud Computing*, *IEEE Communications Magazine*, *IEEE Network*, the *IEEE Consumer Electronics Magazine*, *Future Generation Computing Systems*, *Journal of Network and Computer Applications*, the *Information Sciences*, *Computer Networks*, *Computer and Security*, and the *Journal of Parallel and Distributed Computing* and top-tier conferences such as ACM MobiHoc, IEEE Globecom, IEEE ICC, IEEE WiMob, etc. He has been Guest Editor for Special Issues in IEEE Network, IEEE Transactions on Industrial Informatics, Computer Communications, Elsevier, Software: Practice and Experience, Wiley, Transactions on Emerging Telecommunications Technologies, Wiley, Security and Privacy, Wiley. He has been Workshop Chair for many conferences including IEEE Infocom, IEEE Globecom, IEEE ICC and IEEE PiCom. He is a Senior member of the IEEE and a member of the ACM.



Neeraj Kumar (M'16, SM'17) received the MTech degree from Kurukshetra University, India, and the PhD degree in CSE from SMVD University, Katra. He is an Adjunct Professor with the King Abdulaziz University, Zeddah, Saudi Arabia. He is a full professor with the Department of Computer Science and Engineering, Thapar Institute of Engineering & Technology. He was a postdoctoral research fellow with Coventry University, United Kingdom. He has more than 400 research papers in leading journals and conferences of repute such as the *IEEE Transactions on Knowledge & Data Engineering*, *IEEE Transactions on Industrial Informatics*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Vehicular Technology*, the *IEEE Transactions on Cloud Computing*, *IEEE Internet of Things Journal*, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Smart Grid*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Consumer Electronics*, *IEEE Transactions on Emerging Topics Computing*, *IEEE Transactions on Multimedia*, and *IEEE Transactions on Network and Service Management*. *IEEE Communications Magazine*, *IEEE Network*, the *IEEE Wireless Communications*, *IEEE Consumer Electronics Magazine*, *Future Generation Computing Systems*, *Journal of Parallel and Distributed Computing*, the *Information Sciences*, and *Computer Networks*. He is an associate editor/technical editor of the *IEEE Communications Magazine*, the *IEEE Network Magazine*, the *International Journal of Communication Systems* (Wiley), the *Journal of Network and Computer Applications* (Elsevier), *ACM Computing Survey* and the *Security and Communications Journal* (Wiley). He has won many research awards from different organizations. He is a senior member of the IEEE.



Albert Y. Zomaya (M'90, SM'97, F'04) is a chair professor of high performance computing & networking with the School of Computer Science, The University of Sydney. He also serves as the director of the Centre for Distributed and High Performance Computing. Professor Zomaya was an Australian Research Council Professorial Fellow during 2010–2014 and held the CISCO Systems Chair Professor of Internetworking during the period 2002–2007 and also was Head of school for 2006–2007 in the same school. Prior to his current appointment he

was a Full Professor in the Electrical and Electronic Engineering Department at the University of Western Australia, where he also led the Parallel Computing Research Laboratory during the period 1990–2002. He served as Associate—, Deputy—, and Acting—Head in the same department, and held numerous visiting positions and has extensive industry involvement. Professor Zomaya published more than 550 scientific papers and articles and is author, co-author or editor of more than 20 books. He served as the Editor in Chief of the *IEEE Transactions on Computers* (2011–2014). Currently, Professor Zomaya serves as a Founding Editor in Chief of the *IEEE Transactions on Sustainable Computing*, Founding Co-Editor in Chief of the *IET Cyber-Physical Systems*, and Associate Editor-in-Chief (Special Issues), *Journal of Parallel and Distributed Computing*. He also serves as associate editor for 22 leading journals, such as, the *ACM Computing Surveys*, *ACM Transactions on Internet Technology*, and *IEEE Transactions on Cloud Computing*. Professor Zomaya is the Founding Editor of several book series, such as, the *Wiley Book Series on Parallel and Distributed Computing*, *Springer Scalable Computing and Communications*, and the *IET Book Series on Big Data*. Professor Zomaya was the Chair of the IEEE Technical Committee on Parallel Processing (1999–2003) and currently serves on its executive committee. He is Co-Chair of the IEEE Technical Committee on Cyber-Physical Systems (IEEE Systems Council), Communication Chair of the IEEE Special Technical Community on Sustainable Computing (IEEE Computer Society), Founding Member of the Smart World Technical Committee (IEEE Computational Intelligence Society), and Founding Member of the Technical Committee on Big Data (IEEE Communications Society). Professor Zomaya is a Fellow of the IEEE, the American Association for the Advancement of Science, the Institution of Engineering and Technology (UK). He received the 1997 Edgeworth David Medal from the Royal Society of New South Wales for outstanding contributions to Australian Science. Professor Zomaya is the recipient of the IEEE Technical Committee on Parallel Processing Outstanding Service Award (2011), IEEE Technical Committee on Scalable Computing Medal for Excellence in Scalable Computing (2011), IEEE Computer Society Technical Achievement Award (2014), ACM MSWIM Reginald A. Fessenden Award (2017), Premier Prizes for Science and Engineering for Excellence in Engineering or Information and Communications Technology (2019) and many best paper awards.